Jar Impact Model - Equation Summary

Tasman Jar Placement Simulator | Transparent physical model summary

> This document describes the current open model used by the simulator. It is not the proprietary NOV / Bowen impact engine. The purpose is to make every assumption visible so the model can be calibrated against real reports and field experience.

1. What the Model Tries to Estimate

The model estimates upward jarring impact using an energy balance. Pulling on the work string stores elastic energy. If an accelerator is installed, it adds additional stored energy. Only part of that energy reaches the jar/fish because survey geometry, side contact, clearance, mud damping, mass coupling, and spacing losses reduce the delivered energy.

```
Delivered Energy = Stored Energy x loss/efficiency factors
```

```
Impact Force ~= Pull Load + Delivered Energy / Effective Stop Distance
```

2. Steel Cross-Sectional Area

```
A = (pi / 4) x (OD^2 - ID^2)
```

Term	Meaning
A	Metal cross-sectional area of the work string, in^2.
OD, ID	Work string outside and inside diameters, inches.

Purpose: area controls axial stiffness. Smaller area means more stretch under the same pull load. In the code, a minimum effective area is applied to avoid unrealistically high impact from very small strings.

3. Axial Stretch of the Work String

```
delta_string = F x L x 12 / (A x E)
```

Term	Meaning
delta_string	Elastic stretch in the work string, inches.
F	Applied pull load, lbf.
L	Work string length from surface to top of BHA, ft.
A	Metal area, in^2.
E	Steel modulus, assumed 30,000,000 psi.

Purpose: estimates how much elastic displacement is available to release energy when the jar fires.

4. Elastic Energy Stored in the Work String

```
E_string = 0.5 x F x delta_string / 12
```

Term	Meaning
E_string	Elastic energy stored in the work string, ft-lbf.
F	Applied pull load, lbf.
delta_string / 12	Stretch converted from inches to feet.

Purpose: converts string stretch into stored energy. This is the main energy source when no accelerator is installed.

5. Energy Stored in the Accelerator / Energizer

```
E_acc = 0.5 x F x stroke_acc / 12
```

Term	Meaning
E_acc	Approximate accelerator stored energy, ft-lbf.
stroke_acc	Accelerator stroke from the datasheet, inches.

Purpose: approximates the accelerator as a spring-like energy storage device. This is a first-order approximation because we do not yet have the real force-vs-stroke curve.

6. Total Stored Energy

```
E_stored = E_string + E_acc
```

Purpose: combines the string and accelerator energy before losses.

7. Survey / Contact Efficiency

```
contact_index =
average(sin(inclination)) +
0.7 x max(MD / TVD - 1, 0) +
0.025 x average(DLS) +
0.010 x max(DLS)
```

```
eta_survey = exp(-mu x contact_index)
```

Term	Meaning
eta_survey	Efficiency factor for survey/contact losses.
mu	Friction coefficient entered by the user.
Inclination	Survey inclination at each station.
MD/TVD	Tortuosity proxy. Higher MD for same TVD implies more directional path.
DLS	Dogleg severity from the survey.

Purpose: penalizes energy transfer in deviated/tortuous wells. A more deviated well generally creates more side contact and loses more energy before the jar/fish.

8. Clearance Efficiency from Hole / Casing ID

```
radial_clearance = (Hole_ID - max_string_OD) / 2
```

```
clearance_ratio = radial_clearance / Hole_ID
```

```
eta_clearance = 0.35 + 0.65 x min(clearance_ratio / 0.12, 1)
```

Term	Meaning
Hole_ID	Hole or casing inside diameter, inches.
max_string_OD	Largest OD in work string/BHA/fish path considered by the model.
eta_clearance	Efficiency factor for tight hole/casing clearance.

Purpose: tight clearance increases contact and damping. In your example report, Hole/Casing ID is 6 in and several components are near 5 in OD, so the clearance factor can materially reduce delivered energy.

9. Mud / Fluid Efficiency

```
eta_fluid = clamp(1 - 0.01 x (MW - 8.4), 0.85, 1.05)
```

Term	Meaning
MW	Mud weight, lb/gal.
eta_fluid	Fluid damping efficiency.

Purpose: approximates damping from heavier fluid. This effect is currently mild and bounded.

10. Mass Coupling from Joints Above Jar

```
W_moving = N_joints x L_joint x w_joint + W_acc
```

```
W_ref = 6 x L_joint x w_joint
```

```
eta_mass = 0.55 + 0.35 x (1 - exp(-W_moving / W_ref))
```

Term	Meaning
W_moving	Moving weight between/above jar, approximated from drill collar joints and accelerator weight.
N_joints	Number of drill collar joints above the jar.
L_joint	Assumed joint length, 30 ft.
w_joint	Average drill collar weight, lb/ft, from DC rows above the jar.

eta_mass	Mass coupling efficiency.

Purpose: adding drill collar mass helps transfer accelerator/string energy into the jar impact, but the benefit should saturate. This prevents one extra joint from unrealistically adding huge impact.

11. Fish Resistance Factor

```
W_resisted = W_fish + 0.15 x L_fish x w_joint
```

```
eta_fish = clamp(W_resisted / (W_resisted + W_moving), 0.25, 0.85)
```

Term	Meaning
W_fish	Total fish weight from the fish component table.
L_fish	Total fish length.
eta_fish	Factor representing resistance/inertia at the stuck fish.

Purpose: the fish dimensions and weight matter. A heavier/longer fish changes how much of the jarring energy appears as peak impact at the stuck point.

12. Spacing Efficiency

```
spacing_length = N_joints x L_joint
```

```
eta_spacing = exp(-spacing_length / 1800)
```

Purpose: accounts for energy loss as the distance between jar and accelerator/mass increases. The current decay length is a calibration constant.

13. Delivered Energy

```
E_delivered =
E_stored x eta_survey x eta_clearance x eta_fluid x eta_mass x eta_fish x eta_spacing
```

Purpose: combines all losses into the energy actually available at the impact event.

14. Peak Impact Force

```
Impact = F + E_delivered / d_stop
```

Term	Meaning
Impact	Estimated peak upward impact force, lbf.
F	Applied firing/pull load, lbf.
d_stop	Effective stopping distance, ft. Internally derived from jar stroke with a minimum of 1 in.

Purpose: converts delivered energy into an equivalent peak force over a short stopping distance. This is one of the most sensitive assumptions in the model.

15. Impact Ratio

```
Impact Ratio = Impact / F
```

Purpose: normalizes impact against the applied pull load. A ratio near 3 means the predicted peak force is roughly three times the applied pull load.

16. Impulse

```
m = W_moving / g
```

```
Impulse = sqrt(2 x m x E_delivered)
```

Term	Meaning
Impulse	Momentum transfer estimate, lb-sec.
m	Moving mass, slugs.
g	32.174 ft/s^2.

Purpose: estimates momentum transfer. More moving mass can increase impulse even when peak impact does not increase.

17. Why Your Example Can Produce a Low Ratio

> In the example report reviewed, the model sees a tight 6 in Hole/Casing ID with approximately 5 in fish/BHA OD, friction coefficient 0.40, and a deviated survey. Those three items reduce eta_clearance and eta_survey. If the selected work string is small, the model limits area sensitivity to avoid non-physical impact spikes. Together, these assumptions can reduce delivered energy enough to produce an impact ratio near 3.

18. Calibration Items

The following constants should be calibrated against trusted NOV/Bowen/field reports:

Constant / Assumption	Current Use	Why It Matters
Minimum effective work string area	Prevents runaway stretch for very small strings.	Controls sensitivity to OD/ID.
Survey contact index coefficients	Inclination, tortuosity and DLS weighting.	Controls losses in deviated wells.
Clearance efficiency curve	Uses radial clearance relative to hole ID.	Controls losses in tight casing/hole.
Mass coupling curve	Saturating benefit from DC joints above jar.	Controls impact change when adding collars.
Spacing decay length	Penalizes long spacing from jar to mass/accelerator.	Controls effect of separation.

Effective stop distance	Derived from jar stroke with a minimum.	Strongly controls peak impact.

Generated as a model documentation aid for Tasman Jar Placement Simulator.